

EXPERIMENT - 2

Date: 2022-05-18

TO DETERMINE THE VOLUME AND DENSITY OF GIVEN BALL BY USING MICROMETER SCREW GAUGE

APPARATUS REQUIRED:

- i. Micrometer screw gauge
- ii. Experimental Ball
- iii. Physical Balance

THEORY:

The pitch of the screw is defined as the linear distance travelled by the screw in one complete rotation of the circular scale. It is denoted by the letter P. The least count (L.C.) is the distance travelled by screw when the circular scale is rotated through one circular scale division.

$$\text{L.C.} = \frac{\text{Pitch (P)}}{\text{No. of circular scale divisions (N)}}$$

Volume of the ball = $\frac{\pi d^3}{6}$, where d is the diameter of given ball.

$$\therefore \text{Density } (\rho) = \frac{\text{Mass (M)}}{\text{Volume (V)}}$$

OBSERVATIONS:

No. of circular scale divisions (N) = 100

Value of 10 smallest divisions of the main scale = 10 mm

$\therefore$ Value of 1 smallest division of the main scale = 1 mm

In 4 complete rotations, circular scale moves through 4 divisions of the main scale.

$\therefore$ In 1 complete rotation, circular scale moves through 1 division of the main scale.

$\therefore$ Pitch (P) = 1 mm

Now, Least Count (L.C.) = $\frac{P}{N}$

= 0.01 mm

Zero instrumental error = -ve type (+ve or -ve)

$\therefore$ Zero instrumental error = -7 div = $100 - 93 \times 0.01$ mm (L.C.) = -0.07 mm

Zero Correction = +0.07 mm (-ve or +ve)

Mass of the ball (m) = 1.04 g

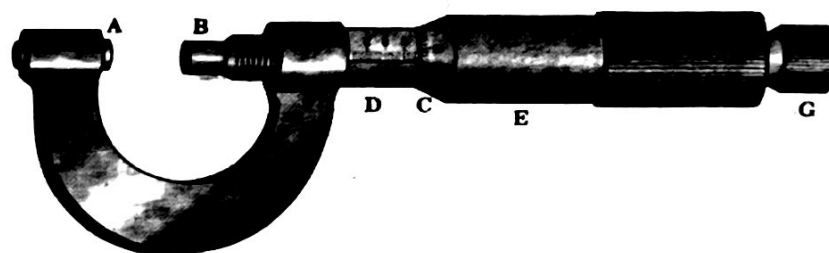
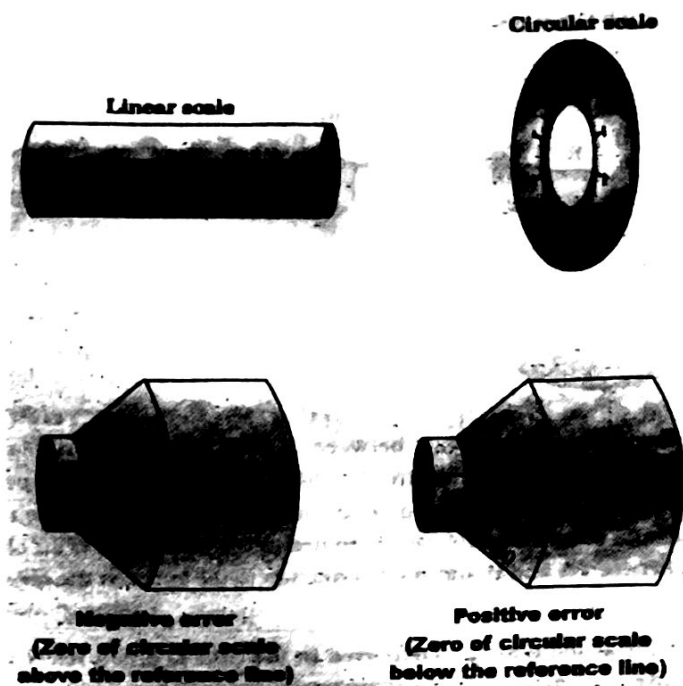


Fig. Micrometer screw gauge

OBSERVATION TABLE:

For diameter of ball:

No. of obs.	Main scale reading (x) mm	Circular scale reading (y) divs.	Value of y = y × L.C. (mm)	Diameter (d') = x + (y × L.C.) (mm)	Apparent diameter (d') mm	Corrected diameter (d) = d' ± correction (mm)
1	6	7	0.07	6.07	6.07	= 6.14
2	6	9	0.09	6.09		
3	6	8	0.08	6.08		
4	6	8	0.08	6.08		
5	6	7	0.07	6.07		
6	6	8	0.08	6.08		
7	6	7	0.07	6.07		
8	6	7	0.07	6.07		
9	6	8	0.08	6.08		
10	6	7	0.07	6.07		

CALCULATION:

$$\begin{aligned} \text{Diameter of ball (d)} &= 6.14 \text{ mm} \\ &= 0.614 \text{ cm} \end{aligned}$$

$$\begin{aligned} \text{Volume of the steel ball (V)} &= \frac{\pi}{6} d^3 \\ &= \frac{22}{7} \times (0.614)^3 = 0.121 \text{ cm}^3 \end{aligned}$$

$$\begin{aligned} \text{Density of steel ball, } (\rho) &= \frac{\text{Mass (M)}}{\text{Volume (V)}} = \frac{1.04}{0.121} \\ &= 8.59 \text{ g/cm}^3 \end{aligned}$$

PERCENTAGE ERROR:

For...density of steel ball

$$\text{Standard value of density, } \rho, (\text{S.V.}) = 7.7 \text{ g/cm}^3$$

$$\begin{aligned} \% \text{ error} &= \left| \frac{\text{S.V.} - \text{O.V.}}{\text{S.V.}} \right| \times 100\% \\ &= \left| \frac{7.7 - 8.59}{7.7} \right| \times 100\% = 11.55\% \end{aligned}$$

RESULT:

The volume and density of the steel ball were found to be 0.121 cm^3 and 8.59 g/cm^3 respectively.

CONCLUSION:

Hence from this experiment, we determined volume and density of the given ball.

SOURCES OF ERROR:

- backlash error
- Carelessness in the experiment
- The non-uniform shape of ball
- excessive pressure applied between the screws.
- defects of vision of experiment performer.
- Measurement in the length of rod.

PRECAUTIONS:

1. The zero error should be observed carefully.
2. The screw should not be pressed too hard.
3. Zero-error should be carefully determined.
4. Each observation should be taken carefully.
5. The error due to parallax in reading should be avoided.
6. The screw should always rotate in same direction to avoid backlash error.

VIVA QUESTIONS:

1. What is the principle of screw gauge?
2. Why is the screw gauge commonly called a micrometer screw gauge?
3. What do you mean by the least count of micrometer screw gauge?
4. What do you mean by the pitch of a screw gauge?
5. What do you mean by the zero error of a micrometer screw gauge?
6. How do you distinguish positive or negative zero error in screw gauge?
7. How can you make the screw gauge more sensitive?
8. Can you find the thickness of a paper by using screw gauge? If yes, how?

1. Screw gauge is based on the principle of a screw. It states that the rotation of the circular scale is directly proportional to the linear distance moved by its circular scale.

2. It is called a micrometer screw gauge because it can measure very small dimensions accurately - up to a micrometer (10^{-6} m).

3. The least count is the smallest value that the instrument can measure.

It is given by:

$$\text{Least count} = \frac{\text{Pitch}}{\text{Number of division on circular scale}}$$

4. The pitch is the distance moved by the screw in one complete rotation.

5. If the zero mark of the circular scale does not coincide with the base line of main scale and is either above or below the base line of the main scale then it's said to be zero error of a screw gauge.

6. If the zero of the circular scale is left behind the main scale zero (a reference line) then the instrument has positive error.

If the zero of circular scale crosses the reference line then the instrument has negative error.

7. We can make the screw gauge more sensitive by increasing the amount of force that we used until it clicks three times. While the object is in between the thimble and the lock. Even if the thimble is locked, the spindle can be moved by pressing on it. This will assist in getting the measure-

ments as accurate as possible.

8. Yes, we can find the thickness of a paper using a screw gauge by stacking papers and ~~the~~ calculating their average.

~~10/8/17~~